STUDENT PROJECT GUIDE

Diabetes

Hospital Readmission

Context

You are working with 101,766 inpatient encounters from 71,518 patients in a historical US hospital dataset covering 1999–2008. Each encounter includes a diabetes diagnosis; it need not be the primary reason for admission. You report to the Director of Care Quality of a hospital network.

An early readmission in this project is a recorded return in less than 30 days. The Director wants to prioritize limited follow-up resources at discharge. The prototype estimates an observed readmission outcome; it does not establish that extra follow-up causes fewer readmissions.

Data provided

Use the course files supplied with this guide as the project dataset. The data originates from the UCI Diabetes 130-US Hospitals dataset:

- `diabetic_data.csv` — one row per hospital encounter. `encounter_id` identifies the encounter; `patient_nbr` identifies the patient and can therefore repeat across encounters. The file contains demographics, admission and discharge information, utilization history, laboratory and procedure counts, diagnoses, medications, and the readmission outcome.
- IDS_mapping.csv — lookup values for admission_type_id, discharge_disposition_id, and admission_source_id. The CSV contains three lookup tables separated by blank rows and new header lines, not one rectangular table. Split on the headers or use the three mappings reproduced below before interpreting or filtering IDs.

The complete field dictionary and the three ID mapping tables are included below. Missingness is field-specific; in the raw CSV some missing values are represented by `?`.

The outcome field readmitted takes three values: <30 (readmitted in less than 30 days), >30 (readmitted after 30 days), and NO (not readmitted).

Core task

At the moment of **hospital discharge**, estimate the risk that a patient will be readmitted as an inpatient in **less than 30 days** and use that risk to prioritize additional follow-up.

Data dictionary

diabetic_data.csv — encounter fields

Field	Meaning
`encounter_id`	Unique identifier for a hospital encounter.
`patient_nbr`	Unique patient identifier; a patient may have multiple encounters.
`race`	Recorded values: Caucasian, AfricanAmerican, Asian, Hispanic, Other, and ? for missing. Preserve source spelling; 2,273 raw encounters have ?.
`gender`	Recorded values: Female, Male, Unknown/Invalid. Keep the three Unknown/Invalid encounters as unknown; do not infer gender or report reliable group metrics from three observations.
`age`	Age band using literal source strings [0-10), [10-20), …, [90-100). These are intervals, not exact ages.
`weight`	Weight band in pounds: [0-25), [25-50), …, [175-200), or >200. The value ? occurs in 98,569 encounters (96.86%). Do not cast bands to numeric weights.
`admission_type_id`	Coded admission type; decode using the mapping table below.

Field	Meaning
discharge_disposition_id	Coded discharge disposition; decode using the mapping table below.
admission_source_id	Coded admission source; decode using the mapping table below.
time_in_hospital	Length of stay in days.
payer_code	Categorical payer code; ? in 40,256 encounters. No payer-code lookup is supplied, so retain codes and do not invent expansions.
medical_specialty	Recorded admitting-physician specialty label; ? in 49,949 encounters. Treat as categorical.
num_lab_procedures	Number of laboratory tests performed during the encounter.
num_procedures	Number of non-laboratory procedures performed during the encounter.
num_medications	Number of distinct generic medications administered during the encounter.
number_outpatient	Outpatient visits in the year before the encounter.
number_emergency	Emergency visits in the year before the encounter.
number_inpatient	Inpatient visits in the year before the encounter.
diag_1	Primary diagnosis as a recorded ICD-9 string. Mostly three-digit codes, but decimal diabetes codes (such as 250.83), V/E codes and ? occur. Preserve the raw string before grouping.
diag_2	Secondary diagnosis as a recorded ICD-9 string. Mostly three-digit codes, but decimal diabetes codes (such as 250.83), V/E codes and ? occur. Preserve the raw string before grouping.
diag_3	Additional secondary diagnosis as a recorded ICD-9 string. Mostly three-digit codes, but decimal diabetes codes (such as 250.83), V/E codes and ? occur. Preserve the raw string before grouping.
number_diagnoses	Number of diagnoses entered for the encounter.
max_glu_serum	Serum glucose test result category: >200, >300, Norm, or None when not measured.
A1Cresult	HbA1c result category: >8, >7, Norm, or None when not measured.
change	Whether diabetes medication changed during the encounter; raw coding is Ch or No.
diabetesMed	Whether a diabetes medication was prescribed/administered: Yes or No.
readmitted	Source outcome: <30 for readmission in less than 30 days; >30 for later readmission; NO for no recorded readmission. These labels do not establish complete follow-up outside the source records.

Medication fields

Shared coding: Each medication field records whether the drug was not prescribed (`No`), prescribed with no dosage change (`Steady`), increased (`Up`), or decreased (`Down`) during the encounter. The source CSV header `citoglipton` is retained as written in the dataset.

Field	Meaning
metformin	Medication status. Observed values: No, Steady, Up, Down.
repaglinide	Medication status. Observed values: No, Steady, Up, Down.
nateglinide	Medication status. Observed values: No, Steady, Up, Down.

Field	Meaning
chlorpropamide	Medication status. Observed values: No, Steady, Up, Down.
glimepiride	Medication status. Observed values: No, Steady, Up, Down.
acetohexamide	Medication status. Observed values: No, Steady.
glipizide	Medication status. Observed values: No, Steady, Up, Down.
glyburide	Medication status. Observed values: No, Steady, Up, Down.
tolbutamide	Medication status. Observed values: No, Steady.
pioglitazone	Medication status. Observed values: No, Steady, Up, Down.
rosiglitazone	Medication status. Observed values: No, Steady, Up, Down.
acarbose	Medication status. Observed values: No, Steady, Up, Down.
miglitol	Medication status. Observed values: No, Steady, Down, Up.
troglitazone	Medication status. Observed values: No, Steady.
tolazamide	Medication status. Observed values: No, Steady, Up.
examide	Medication status. Observed values: No. Constant in this file.
citoglipton	Medication status. Observed values: No. Constant in this file.
insulin	Medication status. Observed values: No, Steady, Down, Up.
glyburide-metformin	Medication status. Observed values: No, Steady, Up, Down.
glipizide-metformin	Medication status. Observed values: No, Steady.
glimepiride-pioglitazone	Medication status. Observed values: No, Steady.
metformin-rosiglitazone	Medication status. Observed values: No, Steady.
metformin-pioglitazone	Medication status. Observed values: No, Steady.

IDS_mapping.csv — admission_type_id

Code	Meaning
1	Emergency
2	Urgent
3	Elective
4	Newborn
5	Not Available
6	NULL

Code	Meaning
7	Trauma Center
8	Not Mapped

IDS_mapping.csv — discharge_disposition_id

Code	Meaning
1	Discharged to home
2	Discharged/transferred to another short-term hospital
3	Discharged/transferred to skilled nursing facility (SNF)
4	Discharged/transferred to intermediate care facility (ICF)
5	Discharged/transferred to another inpatient care institution
6	Discharged/transferred to home with home health service
7	Left against medical advice (AMA)
8	Discharged/transferred home under care of Home IV provider
9	Admitted as an inpatient to this hospital
10	Neonate discharged to another hospital for neonatal aftercare
11	Expired
12	Still patient or expected to return for outpatient services
13	Hospice / home
14	Hospice / medical facility
15	Discharged/transferred within institution to Medicare-approved swing bed
16	Discharged/transferred/referred to another institution for outpatient services
17	Discharged/transferred/referred to this institution for outpatient services
18	NULL
19	Expired at home; Medicaid-only hospice
20	Expired in a medical facility; Medicaid-only hospice
21	Expired, place unknown; Medicaid-only hospice
22	Discharged/transferred to rehabilitation facility or rehabilitation unit
23	Discharged/transferred to long-term care hospital
24	Discharged/transferred to Medicaid-certified nursing facility not certified under Medicare
25	Not Mapped
26	Unknown/Invalid
27	Discharged/transferred to federal health care facility
28	Discharged/transferred/referred to psychiatric hospital/unit
29	Discharged/transferred to Critical Access Hospital (CAH)
30	Discharged/transferred to other health-care institution not defined elsewhere

IDS_mapping.csv — admission_source_id

Code	Meaning
1	Physician Referral
2	Clinic Referral
3	HMO Referral
4	Transfer from a hospital
5	Transfer from a Skilled Nursing Facility (SNF)
6	Transfer from another health care facility
7	Emergency Room
8	Court/Law Enforcement

Code	Meaning
9	Not Available
10	Transfer from critical access hospital
11	Normal Delivery
12	Premature Delivery
13	Sick Baby
14	Extramural Birth
15	Not Available
17	NULL
18	Transfer from another Home Health Agency
19	Readmission to same Home Health Agency
20	Not Mapped
21	Unknown/Invalid
22	Transfer from hospital inpatient/same facility resulting in a separate claim
23	Born inside this hospital
24	Born outside this hospital
25	Transfer from Ambulatory Surgery Center
26	Transfer from Hospice

ICD-9 diagnosis grouping used in Strack et al.

Group	ICD-9 codes	Meaning
Circulatory	390-459, 785	Diseases of the circulatory system
Respiratory	460-519, 786	Diseases of the respiratory system
Digestive	520-579, 787	Diseases of the digestive system
Diabetes	250 and 250.xx	Diabetes mellitus
Injury	800-999	Injury and poisoning
Musculoskeletal	710-739	Musculoskeletal/connective-tissue disease
Genitourinary	580-629, 788	Diseases of the genitourinary system
Neoplasms	140-239	Neoplasms
Other	All remaining ICD-9 codes	All diagnosis codes outside the groups above

Provenance

- **UCI — Diabetes 130-US Hospitals for Years 1999-2008:** https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008
- **Strack et al. (2014), DOI:** https://doi.org/10.1155/2014/781670